Computer Vision ? Homework 1

Task 0: Package versions
NumPy: 2.5.2
Matplotlib: 3.11.1
OpenCV: 5.0.0

Task 1: Image analysis
Image dtype: uint8
Pixel dtype: uint8
Dimensions (height, width, channels): (321, 433, 3)

Task 10: Moments of glasses_outline.png
M00 = 3006960
M10 = 1432355910
M01 = 575000265
M20 = 717459250950
M02 = 125088054195
M11 = 288613029000
x_bar = 476.346845319
y_bar = 191.223117368
mu20 = 35161031847.7
mu02 = 15134711034.4
mu11 = 14713466709.7

Task 11: Orientation and eccentricity
Orientation (clockwise degrees): 27.8814963047
Eccentricity: 0.91040788693

Original image



Task 2: Red image



Task 3: Photographic negative



Task 4: Red / blue channel swap



Task 5: Foliage detection



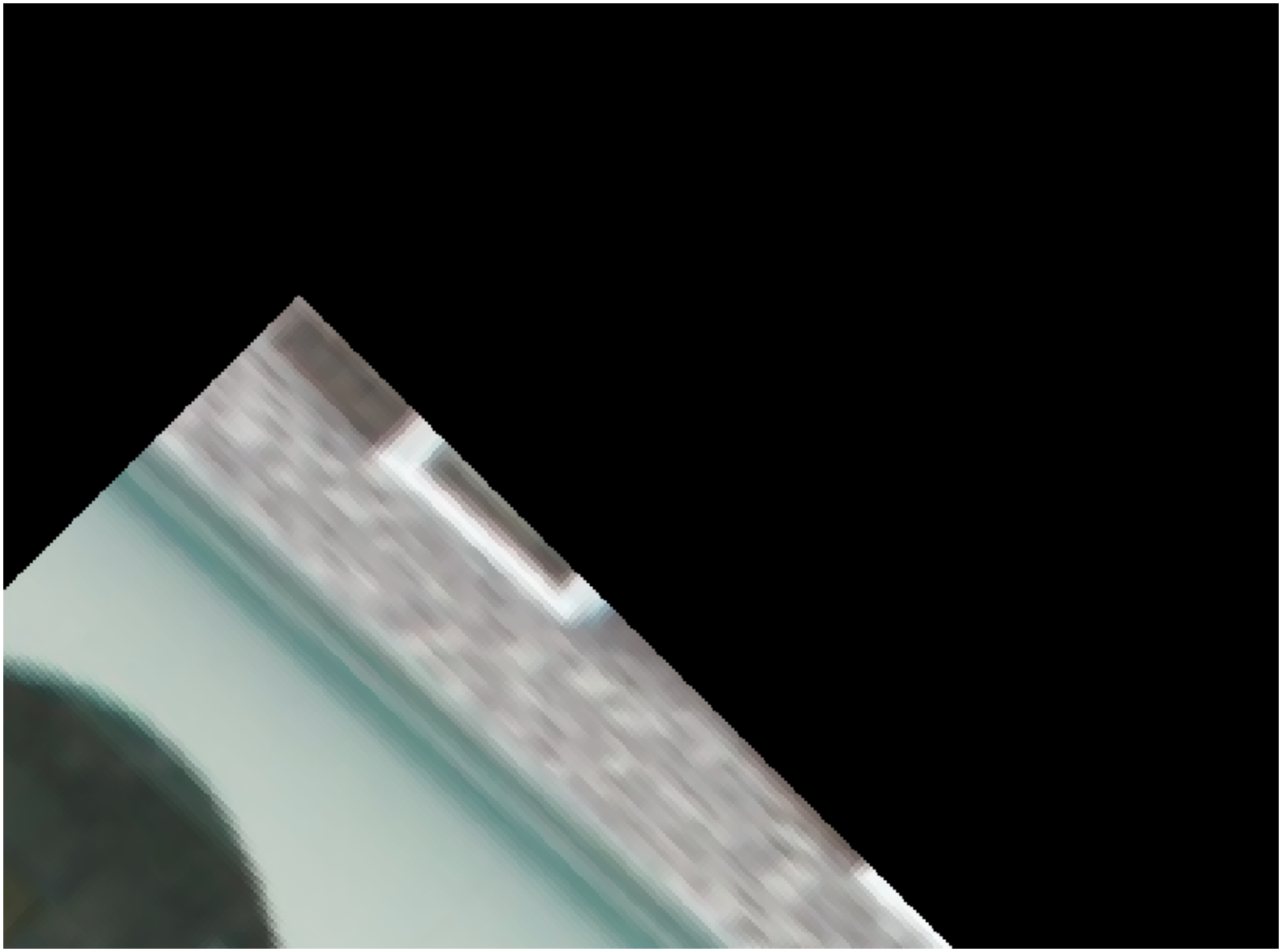
Task 6: Shift right 200, down 100



Task 7: Rotate clockwise 90 degrees



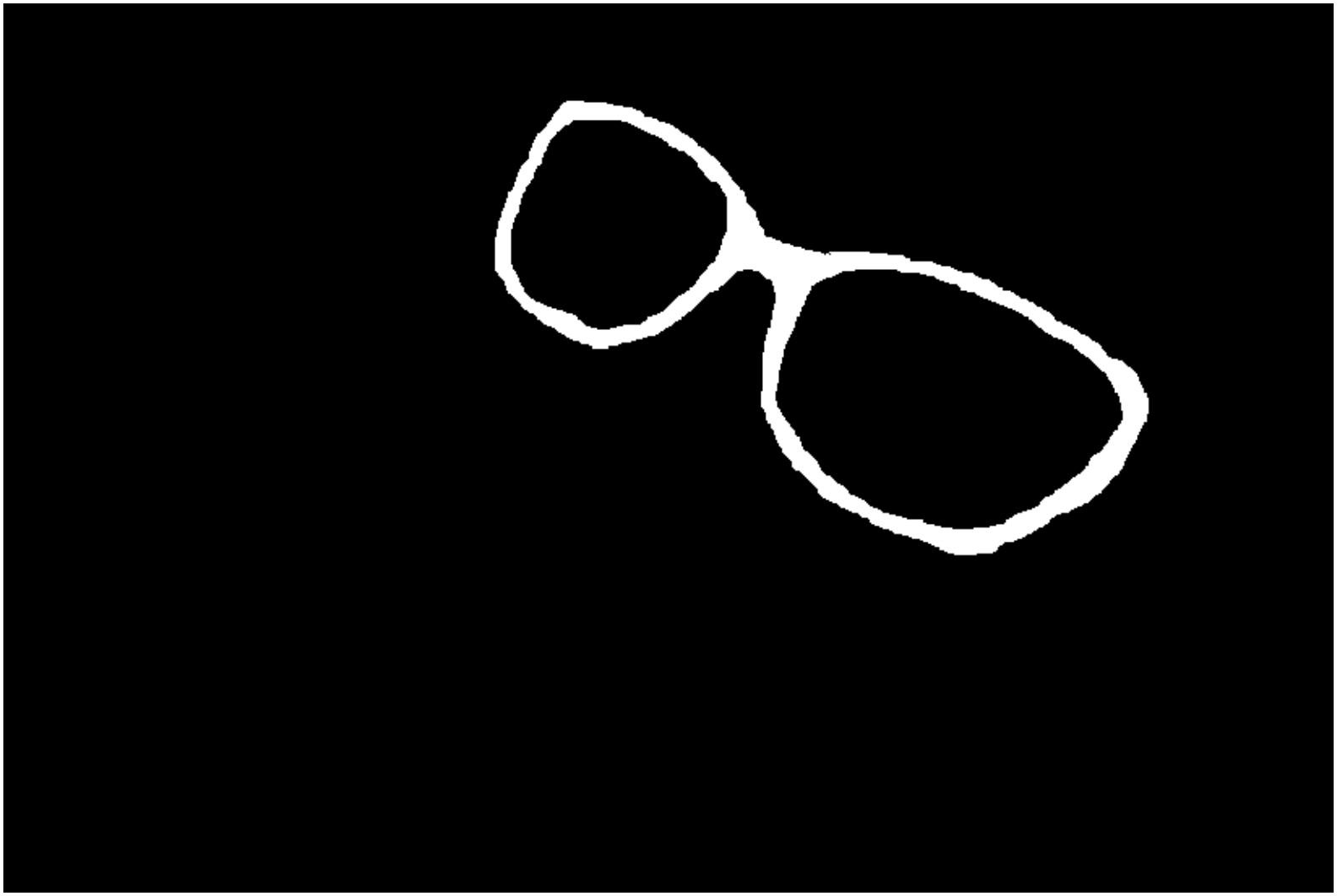
Task 8: Similarity transform ? scale 2, clockwise 45?, shift [100, 100]



Task 9: Grayscale conversion



Task 10: Glasses outline



Task 11: Best-fitting ellipse

